

# **SOFTWARE REQUIREMENTS SPECIFICATION (SRS)**

**YEN 1.0**

**Youth Entrepreneur Network**

Empowering Young Founders with Knowledge, Capital & Connections

**Document Version**

Version 1.0

**Prepared By**

Sanjana B

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## 1.Introduction

Youth Entrepreneur Network (YEN) is a web-based digital entrepreneurship ecosystem designed to guide students and young founders through the entire startup journey. The platform enables users to create startup profiles, access mentorship, join incubator programs, pitch to investors, attend demo days, and collaborate with peers—all in one centralized system.

YEN eliminates common barriers faced by emerging entrepreneurs such as lack of mentorship, limited funding access, and disconnected resources. By offering a unified and user-friendly interface, it helps founders manage their progress from idea to funding in a structured and transparent environment. The platform supports founders, mentors, investors, and incubators, ensuring opportunities are based on merit, traction, and execution.

### 1.1Purpose

This Software Requirements Specification (SRS) document outlines the functional and non-functional requirements of Youth Entrepreneur Network (YEN) 1.0. It serves as a reference for:

- **Developers** – for system design, implementation, and enhancements
- **Colleges & Incubators** – to understand program integration and support workflows
- **Mentors & Investors** – to evaluate startup discovery and engagement processes
- **Founders** – to understand platform usage and startup management
- **System Administrators** – for governance, monitoring, and platform operations

### 1.2Scope

This document applies to the **Youth Entrepreneur Network (YEN)** software system. YEN is a centralized web-based platform that enables students and young founders to digitally manage their complete startup journey in a structured and professional environment. The platform allows founders to create and manage startup profiles, track progress, receive mentorship, participate in incubator programs, submit pitches to investors, and apply for funding and demo day opportunities.

The system accepts inputs from multiple types of users including founders, mentors, investors, and incubators. These inputs include personal and startup profile data, mentorship requests, learning activity, pitch decks, funding applications, and program participation details. The system processes this information to generate meaningful outputs such as role-based dashboards, startup performance summaries, mentorship schedules, investor pitch views, incubator cohort tracking, and real-time communication updates.

The YEN platform is designed to be cloud-hosted, secure, and highly scalable. It is built to support future expansion including advanced analytics, startup recommendation engines, and AI-driven matching between founders, mentors, and investors, ensuring long-term growth and adaptability of the ecosystem.

## 1.2 Definitions, Acronyms, and Abbreviations

| TERM             | DEFINITION                                         |
|------------------|----------------------------------------------------|
| <b>YEN</b>       | Youth Entrepreneur Network                         |
| <b>Founder</b>   | A student or young entrepreneur building a startup |
| <b>Mentor</b>    | An experienced professional guiding founders       |
| <b>Investor</b>  | An individual or firm that funds startups          |
| <b>Incubator</b> | An organization that runs startup programs         |
| <b>UI</b>        | User Interface                                     |
| <b>UX</b>        | User Experience                                    |
| <b>API</b>       | Application Programming Interface                  |
| <b>SaaS</b>      | Software as a Service                              |
| <b>DBMS</b>      | Database Management System                         |
| <b>SRS</b>       | Software Requirements Specification                |

|                   |                                           |
|-------------------|-------------------------------------------|
| <b>MVP</b>        | Minimum Viable Product                    |
| <b>Pitch Deck</b> | A presentation used to pitch a startup    |
| <b>Demo Day</b>   | Event where startups present to investors |

### 1.3 References

The references for the **Youth Entrepreneur Network (YEN)** software are as follows:

- [1]** IEEE Software Requirements Specification Standard (IEEE 830 / IEEE 29148) – Guidelines for preparing structured software requirement documents.
- [2]** Startup India – Entrepreneurship Development Framework – Government initiatives supporting startups and incubation.
- [3]** Y Combinator Startup Playbook – Standard startup building, validation, and funding practices.
- [4]** AngelList Startup Platform – Reference for startup discovery and investor matching.
- [5]** NASSCOM Startup Ecosystem Reports – Industry insights on technology startups and entrepreneurship.

### 1.4 Overview

Section **1.0** discusses the purpose, scope, definitions, and references related to the **Youth Entrepreneur Network (YEN)** software.

Section **2.0** describes the overall product perspective, system functionalities, operating environment, constraints, and user characteristics of the platform.

Section **3.0** details the functional and non-functional requirements necessary for the design and implementation of the YEN software.

## 2. Overall Description

### 2.1 Product Perspective

- **Youth Entrepreneur Network (YEN) 1.0** is a cloud-based web platform composed of multiple integrated software modules designed to support student founders, mentors, investors, incubators, and startup institutions.
- The software enables young entrepreneurs to manage their startup journey digitally by creating startup profiles, tracking progress, receiving mentorship, accessing learning resources, and connecting with investors—without relying on informal networks or fragmented tools.
- In addition to founder self-management, the system supports operations such as mentor discovery, investor matchmaking, incubator cohort management, demo days, and startup visibility.
- The platform consists of core modules including User & Role Management, Startup Profile Management, Learning & Skill Development, Mentorship System, Funding & Investor Discovery, Incubator & Cohort Management, Events & Demo Day System, Messaging & Collaboration, and Administration & Analytics.
- YEN communicates with cloud-based servers through the internet to store, process, and synchronize startup data, user accounts, pitch materials, mentoring sessions, and investor interactions.
- The system is designed to be scalable and future-ready, with provisions for AI-based startup evaluation, automated mentor and investor matching, and data-driven recommendations in later phases.
- The minimum system resources on client devices shall be sufficient to support smooth web-based interaction, secure authentication, and real-time data access.
- The platform shall support thousands of founders, mentors, investors, and incubators concurrently while maintaining high performance, data security, and system reliability.

## **2.2 Product Function:**

### **Startup & Founder Management**

- Founders can create detailed personal and startup profiles
- Each startup has its own digital identity
- Startups can upload pitch decks, team details, traction, and progress

### **Learning & Skill Development**

- Founders can access startup courses and playbooks
- Progress tracking and certificates are supported
- Live and recorded workshops are available

### **Mentorship System**

- Mentors create verified profiles
- Founders can request mentorship sessions
- One-to-one and group mentoring is supported

### **Investor & Funding System**

- Investors can browse startups
- Founders can submit pitches
- Demo days and funding programs are managed

### **Incubator & Cohort Management**

- Incubators create startup cohorts
- They assign mentors and track progress
- They organize bootcamps and demo days

### **Communication & Collaboration**

- In-app messaging
- Team collaboration
- File sharing and updates

### **Administration & Governance**

- User verification and approval
- Content moderation
- Platform analytics and system control

## **2.3 User Characteristics**

### **User A – Founder / Student Entrepreneur**

This user is typically a college student or young individual starting a business. They may have limited exposure to professional startup ecosystems. The platform provides simple onboarding, guided startup creation, and step-by-step access to learning, mentors, and funding opportunities. The interface is designed to be intuitive, beginner-friendly, and mobile-accessible.

### **User B – Mentor**

This user is an experienced professional, entrepreneur, or industry expert. They use the system to review startups, provide guidance, schedule mentoring sessions, and communicate with founders. Their interface focuses on viewing startup data, mentoring requests, and communication tools.

### **User C – Investor**

This user is an angel investor, VC, or funding organization. They use YEN to discover startups, view pitch decks, filter startups by industry and traction, and participate in demo days. Their interface prioritizes startup discovery and evaluation.

### **User D – Incubator / College**

This user represents an institution running startup programs. They use the system to manage cohorts, enroll startups, assign mentors, run events, and track outcomes.

### **User E – Administrator**

This user manages the entire platform. They approve accounts, verify startups, moderate content, manage funding programs, and ensure platform integrity.

## **2.4 Constraints**

- The platform requires internet connectivity for all real-time operations
- Only registered users can access platform features
- Role-based access must be enforced
- Startup and financial data must be securely stored
- Only verified mentors and investors can interact with startups
- The platform must be accessible from standard web browsers

## **2.5 Assumptions and Dependencies**

- Users have access to a web-enabled device
- Mentors and investors participate voluntarily
- Cloud services and hosting providers remain available
- Email, notifications, and video services remain operational
- Future AI features depend on availability of data

# **3. External Interface Requirements**

## **3.1 User Interface Requirements**

The Youth Entrepreneur Network (YEN) platform shall provide a modern, professional, and highly intuitive user interface designed specifically for students, founders, mentors, investors, and incubators. The interface shall be web-based and responsive, ensuring smooth operation across desktops, laptops, and mobile browsers.

The UI shall be designed with startup-focused workflows, enabling users to move easily from idea stage to mentorship, incubation, and funding. The platform shall present information in a clear, organized, and visually guided manner so that first-time founders can navigate without confusion.

The following screens and interactions shall be provided:



- **Landing Page** – Displays platform introduction, benefits, and role selection (Founder, Mentor, Investor, Incubator, Admin).
- **Registration & Login Screens** – Allow users to create accounts and authenticate based on selected role.
- **Role-Based Dashboards** – Each user sees a personalized dashboard based on their role.
- **Founder Profile Screen** – Displays personal details, skills, and startup associations.
- **Startup Profile Screen** – Displays startup idea, team, traction, pitch deck, and status.
- **Learning Hub** – Provides access to startup courses, playbooks, and training.
- **Mentorship Screen** – Allows founders to request mentors and schedule sessions.
- **Funding & Pitch Screen** – Enables pitch uploads, demo day applications, and investor discovery.
- **Incubator Dashboard** – Allows incubators to manage cohorts, startups, and mentors.
- **Events & Demo Days Screen** – Displays startup events, pitch competitions, and hackathons.
- **Messaging & Collaboration Screen** – Allows secure communication between users.
- **Admin Panel** – Provides system control, verification, analytics, and moderation.
- **Error and Alert Screens** – Displayed for authentication errors, permission violations, or system issues.

#### **Additional UI Specifications**

- The UI shall support light and dark themes
- The interface shall be fully responsive
- Navigation shall be menu-driven
- Icons and progress indicators shall be used
- Forms shall have inline validation
- Accessibility shall be considered

### **3.2 Hardware Interface Requirements**

YEN is a cloud-based web application and shall run on standard user devices.

The system shall support:

Desktop computers

Laptop computers

Tablets

Smartphones (via web browser)

Users shall require only a modern web browser and internet connection. No special hardware is required for Phase 1.

### **3.3 Software Interface Requirements**

- The YEN platform shall be implemented using a full-stack web architecture with React.js for the frontend, Node.js and Express.js for the backend, and MongoDB for the database.
- RESTful APIs shall be used to enable secure and structured communication between the frontend and backend services.
- The system shall integrate with email services to support user verification, password recovery, and notification delivery.
- Video meeting platforms shall be integrated to enable online mentorship sessions and virtual meetings.
- Cloud-based file storage services shall be used to store and manage pitch decks and startup documents securely.
- Analytics and logging systems shall be integrated to monitor platform usage, performance, and system health.

### **3.4 Communication Interface Requirements**

- The system shall use secure internet-based communication to exchange data between users and cloud servers.
- All API requests and responses shall be transmitted using HTTPS to ensure data security.
- JWT (JSON Web Tokens) shall be used for secure authentication and session management.

- WebSocket or equivalent real-time communication protocols may be used for chat, notifications, and live updates.
- All data transferred between client and server shall be encrypted to prevent unauthorized access.
- Offline caching mechanisms may be used to temporarily store user data when internet connectivity is unavailable and synchronize it later.

## **4. System Features**

### **1. User Registration, Authentication & Role Management**

#### **Description**

The Youth Entrepreneur Network provides a secure and role-based authentication system. Every user must create an account by selecting a role: Founder, Mentor, Investor, Incubator, or Administrator. The system ensures that each user only sees and accesses the features relevant to their role.

When a new user registers, the system collects personal details, verifies email or phone number, and stores credentials in encrypted form. Mentor, Investor, and Incubator accounts require administrative approval before becoming active to ensure trust and quality within the ecosystem.

Once logged in, users are redirected to their role-specific dashboard.

#### **Validity Checks**

- Duplicate emails are not allowed
- Password must meet security rules
- Role must be selected
- Non-founder roles require admin approval

#### **Error Handling**

- Invalid login → error message
- Pending approval → account pending message
- Wrong password → retry limit enforced

## **2. Startup Creation and Digital Identity**

### **Description**

Founders can create a startup profile that acts as the digital identity of their business. This includes startup name, industry, description, team members, traction, and pitch materials. The startup profile is the core unit used by mentors, investors, and incubators to evaluate and engage with founders. Only founders can create or edit startup profiles. Each startup has a status such as Idea, MVP, Growth, or Funded.

### **Validity Checks**

- One founder can own only one startup
- Required fields must be filled
- Files must be in supported formats

### **Error Handling**

- Missing fields → prompt user
- Upload failure → retry message

## **3. Learning & Skill Development**

### **Description**

The platform provides startup education through structured courses, founder playbooks, and workshops. Founders can enroll in learning modules related to ideation, marketing, fundraising, legal, and product development.

Progress is tracked and certificates may be generated.

### **Validity Checks**

- User must be a Founder
- Course enrollment must be recorded

## **4. Mentorship System**

### **Description**

Mentors create profiles listing their expertise and availability. Founders can browse mentors and send mentoring requests. Mentors can accept or decline requests. Once accepted, the session is scheduled and communication tools are enabled.

### **Validity Checks**

- Mentor must be approved
- Session cannot overlap

## **5. Funding & Investor Interaction**

### **Description**

Founders upload pitch decks and apply for funding programs or demo days. Investors can browse startups, view pitches, and shortlist companies.

This creates a transparent and merit-based funding pipeline.

## **6. Incubator & Cohort Management**

### **Description**

Incubators and colleges can create cohorts, enroll startups, assign mentors, and manage demo days and hackathons.

They track startup progress inside their programs.

## **7. Collaboration & Communication**

### **Description**

The platform includes chat, file sharing, and team tools to enable founders, mentors, and incubators to collaborate in one place.

## **4.1 Functional Requirements**

Each requirement defines exactly what the system must do.

### **FR-1 User Registration & Authentication**

- The system shall allow users to register using email and password.
- The system shall require users to select a role (Founder, Mentor, Investor, Incubator, Admin).
- The system shall encrypt all passwords before storing them.
- The system shall authenticate users using JWT-based login.
- The system shall prevent access to protected routes without valid authentication.

### **FR-2 Role-Based Access Control**

- The system shall assign permissions based on user role.
- The system shall restrict founders from accessing admin or investor tools.
- The system shall restrict investors and mentors from editing startup data.

### **FR-3 Startup Profile Management**

- The system shall allow founders to create one startup profile.
- The system shall allow founders to upload pitch decks and team data.
- The system shall store startup profiles in the database.

### **FR-4 Learning & Training**

- The system shall allow founders to enroll in startup courses.
- The system shall track learning progress.
- The system shall display certificates upon completion.

### **FR-5 Mentorship**

- The system shall allow mentors to create profiles.
- The system shall allow founders to request mentorship.
- The system shall allow mentors to accept or decline requests.

#### **FR-6 Investor & Funding**

- The system shall allow founders to submit pitch decks.
- The system shall allow investors to browse startups.
- The system shall allow demo day participation.

#### **FR-7 Incubator Management**

- The system shall allow incubators to create cohorts.
- The system shall allow incubators to assign mentors.
- The system shall allow incubators to track startups.

#### **FR-8 Messaging**

- The system shall allow secure messaging between users.
- The system shall notify users of new messages.

#### **FR-9 Administration**

- The system shall allow admins to approve users.
- The system shall allow admins to moderate content.
- The system shall log all admin actions.

## **4.2 Non-Functional Requirements**

This section describes the quality attributes and constraints that define how the **Youth Entrepreneur Network (YEN) 1.0** platform performs its functions. These requirements ensure reliability, usability, security, and scalability of the system.

### **4.2.1 Performance Requirements**

- The system shall respond to user actions such as page loading, form submission, and navigation within 3 seconds under normal network conditions.
- Startup profile updates, mentor requests, and pitch submissions shall be processed without noticeable delay.

- Real-time interactions such as messaging and notifications shall be delivered with minimal latency.
- The system shall support multiple concurrent users without significant performance degradation.

#### **4.2.2 Reliability Requirements**

- The system shall operate continuously with minimal downtime.
- The system shall prevent data loss during server failures, network interruptions, or application crashes.
- All critical transactions such as startup creation, pitch submission, and mentor requests shall be logged.
- The platform shall recover gracefully from unexpected failures.

#### **4.2.3 Security Requirements**

- The system shall enforce secure authentication using encrypted passwords and JWT-based login.
- All data transmission shall use HTTPS with SSL/TLS encryption.
- Role-based access control shall restrict users from accessing unauthorized data or features.
- Startup data, pitch decks, and financial information shall be protected from unauthorized access.
- Administrative actions shall require elevated authentication and authorization.

#### **4.2.4 Usability Requirements**

- The user interface shall be simple, intuitive, and easy to use for students and first-time founders.
- The platform shall provide guided workflows for onboarding and startup creation.
- All menus, buttons, and dashboards shall be clearly labeled and easy to understand.
- The system shall be fully responsive across desktop and mobile devices.

#### **4.2.5 Compatibility Requirements**

- The platform shall be compatible with modern web browsers including Chrome, Firefox, Safari, and Edge.
- The system shall function on Windows, macOS, Linux, Android, and iOS devices.



- The interface shall adapt to different screen sizes and resolutions.

#### **4.2.6 Scalability Requirements**

- The system shall support growth in the number of users, startups, and stored data.
- The architecture shall allow integration of AI-based recommendation and matching engines in future phases.
- Backend services and database shall scale horizontally as usage increases.

#### **4.2.7 Maintainability Requirements**

- The system shall be modular to allow easy updates and feature enhancements.
- Changes to one module shall not affect other system components.
- Logging and monitoring tools shall be available for troubleshooting and maintenance.

#### **4.2.8 Portability Requirements**

- The system shall be deployable on multiple cloud hosting platforms.
- The platform shall support migration to mobile applications in future releases.
- Minimal configuration changes shall be required when moving between environments.

#### **4.2.9 Availability Requirements**

- The platform shall maintain an availability of at least 99% excluding scheduled maintenance.
- Planned maintenance activities shall be communicated to users in advance.

#### **4.2.10 Legal and Compliance Requirements**

- The system shall comply with applicable data protection and privacy laws.
- User consent shall be obtained before collecting or displaying personal and startup data.
- Data storage and access policies shall follow ethical and regulatory standards.

## 5. System Architecture

### 5.1 Architectural Overview

The Youth Entrepreneur Network (YEN) 1.0 platform follows a **multi-layer cloud-based architecture** designed for scalability, security, and modular expansion.

The system consists of four major layers:

**Client Layer (Frontend)**

**Application Layer (Backend Services)**

**Data Layer (Database & Storage)**

**Integration Layer (External Services & APIs)**

Each layer operates independently but communicates securely through well-defined interfaces.

### 5.2 Client Layer

The Client Layer provides the user-facing interface for all platform users including Founders, Mentors, Investors, Incubators, and Administrators.

#### Responsibilities

- User registration and login
- Startup profile creation and editing
- Learning access
- Mentor booking
- Pitch submission
- Messaging
- Viewing dashboards and analytics

#### Supported Platforms

- Desktop web browsers
- Mobile web browsers
- Tablet browsers

- Progressive Web App (future)

### **5.3 Application Layer**

This layer contains all business logic and system workflows.

#### **Core Modules**

- Authentication & Authorization
- Startup Management
- Learning Management
- Mentorship Management
- Funding & Investor Matching
- Incubator & Cohort Management
- Messaging & Notification
- Administration

#### **Responsibilities**

- Process API requests
- Enforce role-based access
- Validate data
- Manage workflow sequences
- Trigger notifications

### **5.4 Data Layer**

This layer stores all platform data.

#### **Stored Data**

- User accounts
- Startup profiles
- Pitch decks

- Mentor sessions
- Investor interactions
- Incubator programs
- Messages
- System logs

### **Characteristics**

- Cloud-based MongoDB
- Encrypted storage
- Backup and recovery
- Scalable data model

## **5.5 Integration Layer**

This layer connects YEN to external services.

### **Integrations**

- Email & notification service
- Video meeting service
- Cloud file storage
- Analytics tools
- Payment gateways (future)
- AI modules (future)

## **5.6 Data Flow**

- User interacts with frontend
- Request sent to backend API
- Backend validates and processes
- Data stored/retrieved from database

- Response returned to client

## 5.7 Security Architecture

- JWT authentication
- Encrypted passwords
- HTTPS communication
- Role-based access control
- Secure file storage

## 6. Database Design

This section describes the database structure and data management strategy for the **Youth Entrepreneur Network (YEN) 1.0** platform. The database is designed to ensure data integrity, security, scalability, and efficient access to startup, user, and funding-related information.

YEN 1.0 uses a centralized cloud-based database to store user accounts, startup profiles, mentoring sessions, investor interactions, and platform activity logs.

### 6.1 Database Overview

The database is responsible for storing and managing:

- User and authentication details
- Founder, mentor, investor, and incubator profiles
- Startup profiles and business data
- Pitch decks and funding applications
- Mentorship sessions and communication records
- Incubator cohorts and events
- System logs and administrative actions

The database follows a modular and structured design to ensure maintainability and scalability.

6.2 Database Type

- **Database Model:** NoSQL (Document-based)
- **Primary Database:** MongoDB (Cloud-hosted)
- **File Storage:** Cloud object storage for pitch decks and documents

This approach supports high performance, flexibility, and scalability for startup data.

6.3 Entity Descriptions

6.3.1 User Collection

Stores authentication and role-related information.

| Field Name    | Description                                     |
|---------------|-------------------------------------------------|
| user_id       | Unique identifier                               |
| name          | Full name                                       |
| email         | Registered email                                |
| password_hash | Encrypted password                              |
| role          | Founder / Mentor / Investor / Incubator / Admin |
| status        | Active / Pending / Blocked                      |
| created_at    | Account creation date                           |

### 6.3.2 Startup Collection

Stores startup-related information.

| Field Name  | Description           |
|-------------|-----------------------|
| startup_id  | Unique startup ID     |
| founder_id  | Linked user           |
| name        | Startup name          |
| industry    | Business category     |
| stage       | Idea / MVP / Growth   |
| description | Business idea         |
| pitch_url   | Pitch deck link       |
| created_at  | Profile creation date |

### 6.3.3 Mentor Profile Collection

| Field Name   | Description         |
|--------------|---------------------|
| mentor_id    | Unique ID           |
| user_id      | Linked user         |
| expertise    | Areas of expertise  |
| experience   | Years of experience |
| availability | Available slots     |
| status       | Approved / Pending  |

#### 6.3.4 Investor Profile Collection

| Field Name       | Description        |
|------------------|--------------------|
| investor_id      | Unique ID          |
| user_id          | Linked user        |
| investment_focus | Industry focus     |
| ticket_size      | Funding range      |
| status           | Approved / Pending |

#### 6.3.5 Mentorship Session Collection

| Field Name | Description                      |
|------------|----------------------------------|
| session_id | Unique ID                        |
| mentor_id  | Linked mentor                    |
| founder_id | Linked founder                   |
| date       | Session date                     |
| status     | Requested / Approved / Completed |

#### 6.3.6 Pitch & Funding Collection

| Field Name  | Description        |
|-------------|--------------------|
| pitch_id    | Unique ID          |
| startup_id  | Linked startup     |
| investor_id | Receiving investor |



|            |                                 |
|------------|---------------------------------|
| status     | Submitted / Reviewed / Accepted |
| created_at | Submission date                 |

### 6.3.7 Incubator Cohort Collection

| Field Name   | Description       |
|--------------|-------------------|
| cohort_id    | Unique ID         |
| incubator_id | Linked incubator  |
| name         | Program name      |
| startups     | Enrolled startups |
| start_date   | Start date        |
| end_date     | End date          |

### 6.3.8 Messaging Collection

| Field Name  | Description      |
|-------------|------------------|
| message_id  | Unique ID        |
| sender_id   | Sender           |
| receiver_id | Receiver         |
| message     | Message text     |
| timestamp   | Sent time        |
| status      | Delivered / Read |

### 6.3.9 Admin Log Collection

| Field Name | Description        |
|------------|--------------------|
| log_id     | Unique ID          |
| admin_id   | Admin user         |
| action     | Performed action   |
| timestamp  | Action time        |
| remarks    | Additional details |

### 6.4 Data Integrity and Security

- All sensitive data shall be encrypted in the database.
- Role-based access control shall restrict unauthorized data access.
- Admin activity shall be fully logged.
- Regular backups shall prevent data loss.

### 6.5 Data Synchronization

- All user actions are stored in real time on cloud servers.
- Cached data is synchronized when connectivity is restored.
- Conflict resolution ensures data consistency.

### 6.6 Database Scalability

- The database shall support horizontal scaling.
- New collections can be added without disruption.
- Optimized queries ensure fast data retrieval.

## **7. Future Enhancements**

This section outlines the proposed enhancements and extensions planned for future versions of the **Youth Entrepreneur Network (YEN) 1.0** platform. These enhancements aim to strengthen startup success, improve founder–investor engagement, expand learning and funding access, and transform YEN into a full-scale digital startup ecosystem.

### **7.1 AI-Based Startup Evaluation**

Future versions of YEN will integrate Artificial Intelligence models to analyze startup profiles, business models, traction data, and founder activity. This will help generate startup quality scores, investor-readiness indicators, and personalized improvement suggestions for founders.

### **7.2 Smart Mentor–Founder Matching**

AI-driven recommendation systems will be introduced to automatically match founders with the most suitable mentors based on industry, startup stage, challenges, and mentor expertise, improving guidance quality and engagement.

### **7.3 Automated Investor Discovery**

YEN will introduce intelligent investor-matching that recommends startups to investors based on investment focus, stage, and historical interest patterns, increasing funding efficiency and deal flow.

### **7.4 Multilingual Platform Support**

The platform will be expanded to support regional and international languages, enabling students and founders from diverse backgrounds to access learning resources, mentoring, and funding in their native language.

### **7.5 Mobile Application Development**

Dedicated mobile applications for Android and iOS will be developed to provide a faster, more personalized user experience with real-time notifications, chat access, and offline viewing of learning content.

## **7.6 Integrated Payment & Subscription System**

Future releases will include subscription plans for premium features such as advanced mentoring, investor access, incubator programs, and learning tracks. Secure online payment gateways will be integrated.

## **7.7 Advanced Startup Analytics Dashboard**

YEN will introduce analytics dashboards showing startup growth metrics such as traction, mentor feedback, investor interest, pitch success rates, and cohort performance trends.

## **7.8 Government & Grant Program Integration**

The platform will integrate government startup schemes, grants, and innovation challenges so founders can discover and apply for public funding opportunities directly through YEN.

## **7.9 API Access for Incubators and Universities**

Secure APIs will be provided for colleges, incubators, and accelerators to integrate YEN with their internal startup tracking systems, enabling automated cohort management and reporting.

## **7.10 Blockchain-Based Startup Verification**

Blockchain technology may be introduced to ensure tamper-proof records for startup registrations, pitch submissions, funding agreements, and founder credentials.

## **7.11 Global Startup Network Expansion**

Future versions will support global expansion, enabling international founders, mentors, and investors to collaborate across borders with multi-currency funding, cross-border programs, and global demo days.

## 8. Conclusion

The **Youth Entrepreneur Network (YEN) 1.0** platform is designed to solve the major challenges faced by young founders and student entrepreneurs by providing a centralized, secure, and professional digital ecosystem for building startups. By enabling structured startup profiles, mentorship access, learning systems, incubator programs, investor engagement, and pitch-based funding, YEN creates a reliable foundation for startup growth and opportunity discovery.

The platform eliminates the dependence on scattered tools such as WhatsApp groups, Google Forms, emails, and informal networking by replacing them with a unified, data-driven startup infrastructure. Its role-based access, secure communication, and organized startup workflows ensure transparency, accountability, and professional engagement between founders, mentors, investors, and incubators.

This Software Requirements Specification (SRS) document defines the functional and non-functional requirements, external interfaces, system architecture, and database design required for the successful development and deployment of **YEN 1.0**. The platform is built to be scalable, secure, and adaptable to future innovations such as AI-based startup matching, automated funding workflows, and global startup collaboration.

YEN aims to create an inclusive, transparent, and opportunity-driven startup ecosystem where every young entrepreneur—regardless of background, location, or college—has equal access to knowledge, mentorship, funding, and growth.

## Appendix

### Appendix A: Glossary

| Term            | Description                                                                               |
|-----------------|-------------------------------------------------------------------------------------------|
| Founder Profile | A digital identity containing a founder's personal, professional, and startup information |

|                   |                                                                                 |
|-------------------|---------------------------------------------------------------------------------|
| Startup Profile   | A digital representation of a startup including idea, team, stage, and traction |
| Mentor System     | A feature enabling founders to receive guidance from experienced professionals  |
| Opportunity Board | Section listing funding programs, demo days, competitions, and grants           |
| Incubator Cohort  | A structured startup program managed by incubators or colleges                  |
| Role-Based Access | Restriction of platform features based on user roles                            |

## Appendix B: User Roles Summary

| Role                | Description                                                             |
|---------------------|-------------------------------------------------------------------------|
| Founder / Student   | Creates startups, joins learning programs, seeks mentorship and funding |
| Mentor              | Guides founders and provides expert advice                              |
| Investor            | Reviews startups and provides funding                                   |
| Incubator / College | Manages startup programs, cohorts, and demo days                        |
| Administrator       | Manages users, approvals, platform content, and system operations       |

## Appendix C: Assumptions Summary

- Users have access to a basic smartphone or web-enabled device
- Internet connectivity may vary depending on location
- Startup data is entered manually in Phase 1
- Mentors, investors, and incubators voluntarily join the platform
- Advanced automation and AI features are planned for future releases

## Appendix D: Abbreviations

| Abbreviation | Expansion                           |
|--------------|-------------------------------------|
| YEN          | Youth Entrepreneur Network          |
| AI           | Artificial Intelligence             |
| API          | Application Programming Interface   |
| SRS          | Software Requirements Specification |
| UI           | User Interface                      |
| UX           | User Experience                     |
| JWT          | JSON Web Token                      |
| MVP          | Minimum Viable Product              |

## Appendix E: Revision History

| Version | Date | Description                                               |
|---------|------|-----------------------------------------------------------|
| 1.0     | —    | Initial SRS creation for Youth Entrepreneur Network (YEN) |

## **Appendix F: Author Information**

**Project Name:** Youth Entrepreneur Network (YEN) 1.0

**Project Type:** Startup Ecosystem Platform

**Developed By:** Sanjana B

**Organization:** Techno Vanam

**Purpose:** Academic, Startup, and Product Development